What Are Acids and Bases?

Answer key

A. Definitions

1. Brønsted–Lowry acid and base.

> An acid is a proton (H^+) **donor**; a base is a proton (H^+) **acceptor**.

2. Strong vs concentrated.

> **Strong** means the acid ionizes completely in water. **Concentrated** means a large amount of acid is dissolved per volume. A dilute solution of a strong acid and a concentrated solution of a weak acid are both possible.

B. Acids, bases and conjugate pairs

3. Identify acid and base.

> (a) Acid: HNO_3; base: H_2O
>
> (b) Acid: HCl; base: NH_3
>
> (c) Acid: HCO_3^-; base: OH^-

4. Conjugate bases.

> (a) F^- (b) HSO_4^- (c) NH_3 (d) OH^-

5. Conjugate acids.

> (a) NH_4^+ (b) H_2O (c) H_3O^+ (d) HCO_3^-

C. Strong and weak

6. Classify.

> HCl and HNO_3: strong acids. CH_3COOH: weak acid. NaOH: strong base. NH_3: weak base.

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D. Neutralization

7. Neutralization equations.

> (a) $\mathrm{HCl + KOH \longrightarrow KCl + H_2O}$
> (b) $\mathrm{H_2SO_4 + 2\,NaOH \longrightarrow Na_2SO_4 + 2\,H_2O}$
> (c) $\mathrm{2\,HNO_3 + Ca(OH)_2 \longrightarrow Ca(NO_3)_2 + 2\,H_2O}$

E. Challenge

8. Water acting both ways.

> Such a substance is **amphoteric**. Water can accept a proton (becoming $\mathrm{H_3O^+}$) when it meets a stronger acid, and donate a proton (becoming $\mathrm{OH^-}$) when it meets a base such as ammonia.

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